

Research & Working Notes

Group 22 — SafeX Educational Learning Platform · Week 3 · Module: Video Management (Search, Review, Publish, Edit, Delete)

1. Architecture Decision

The browser cannot safely hold the YouTube API key and cannot open a SQL Server connection, so the module follows the standard three-tier pattern:

React (browser) → ASP.NET Core Web API → SQL Server, with the Web API also being the only piece that talks to **YouTube Data API v3**.

2. YouTube Data API v3 — Integration

- Key obtained from Google Cloud Console → APIs & Services → Credentials, restricted to *YouTube Data API v3*.
- Key is stored in **appsettings.json** under **YouTube:ApiKey** and read server-side only. It is never sent to the browser.
- Two-step pattern per search: (1) **search.list** to get video IDs, (2) **videos.list** with those IDs to fetch *contentDetails* and *statistics* in one call.
- **safeSearch=strict** is always set — required for a kids-oriented platform.
- ISO-8601 **duration** (e.g. *PT4M13S*) is parsed into seconds by a small helper on the server before returning to React.

3. Quota Handling

Call	Cost (units)	How often
search.list	100	Once per user query (after 500 ms debounce)
videos.list	1	Once per query, batched with up to 50 IDs
Repeated same query	within 10 min	Served from IMemoryCache

Free tier = **10,000 units/day**. With the pattern above one search costs **101 units**, giving roughly **~99 fresh searches/day**. Once a video is imported, its metadata lives in SQL Server, so library / review / publish screens never call YouTube again.

4. Safety Measures

- SafeSearch strict on every search call.
- Admin manually reviews Educational quality and Copyright before a video can be approved.
- Publishing forces an audience choice (**Kids** or **General**) and a category — no video reaches the Kids panel without an explicit admin decision.

5. Database Design (SQL Server)

Categories(Id UNIQUEIDENTIFIER PK, Name UNIQUE, CreatedAt)

Videos(Id PK, YouTubeId UNIQUE, Title, Description, ThumbnailUrl, ChannelTitle, ViewCount, DurationSeconds, Language, Audience, Status, CategoryId FK, ReviewNotes, EducationalQualityOk, CopyrightOk, IsDeleted, DeletedAt, PublishedAt, CreatedAt, UpdatedAt)

Soft delete = **IsDeleted=1** + **DeletedAt**; restore flips them back; permanent delete = SQL DELETE.

6. References

- YouTube Data API v3 — <https://developers.google.com/youtube/v3/docs>
- Quota costs — https://developers.google.com/youtube/v3/determine_quota_cost
- EF Core with SQL Server — <https://learn.microsoft.com/ef/core/>
- ASP.NET Core Web API — <https://learn.microsoft.com/aspnet/core/web-api/>